Out-of-Sample Reweighting to the 69-Student Pool: Composition and Productivity

Real student demographics; fixed-behavior (composition) channel; 8B house style

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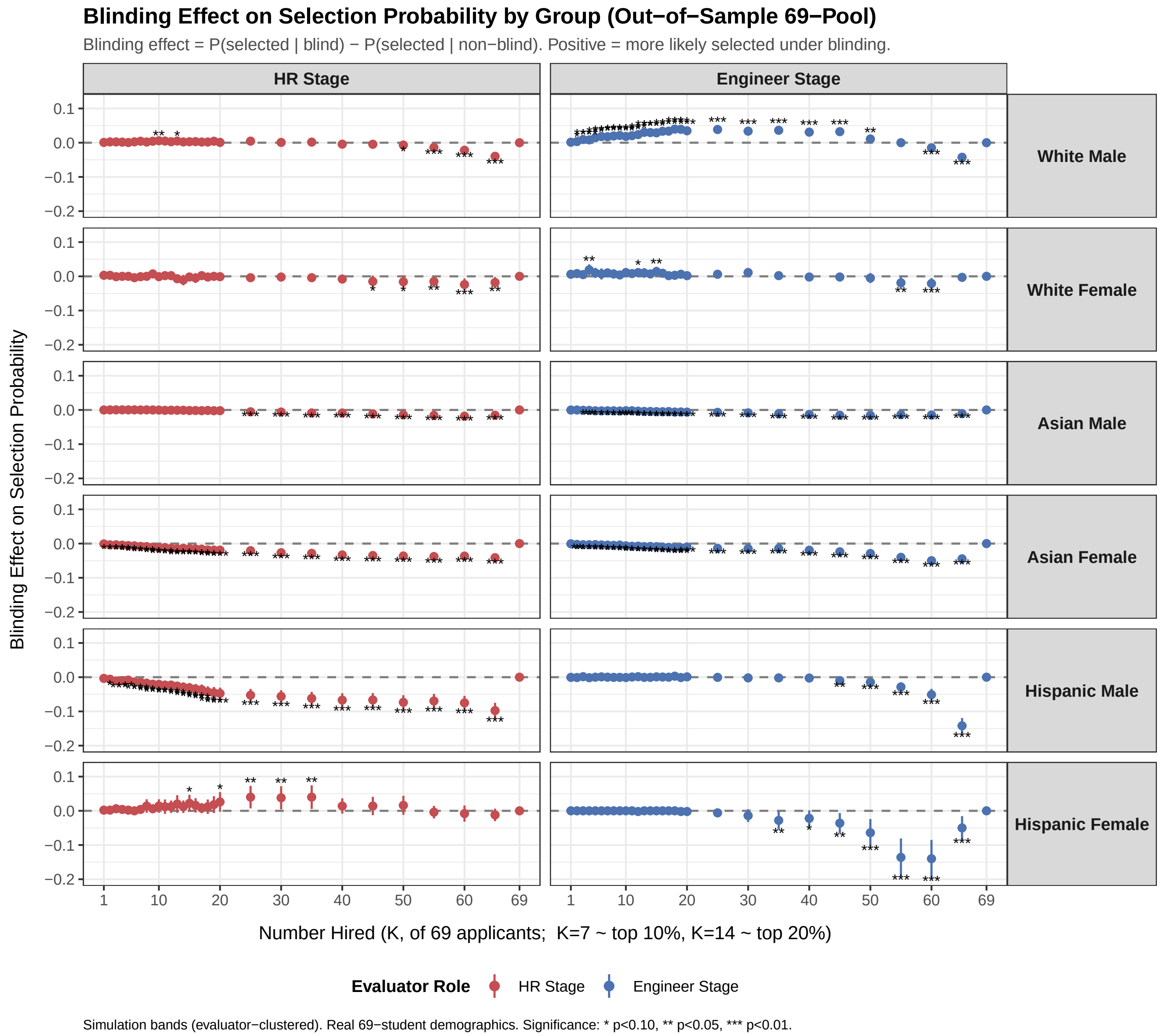
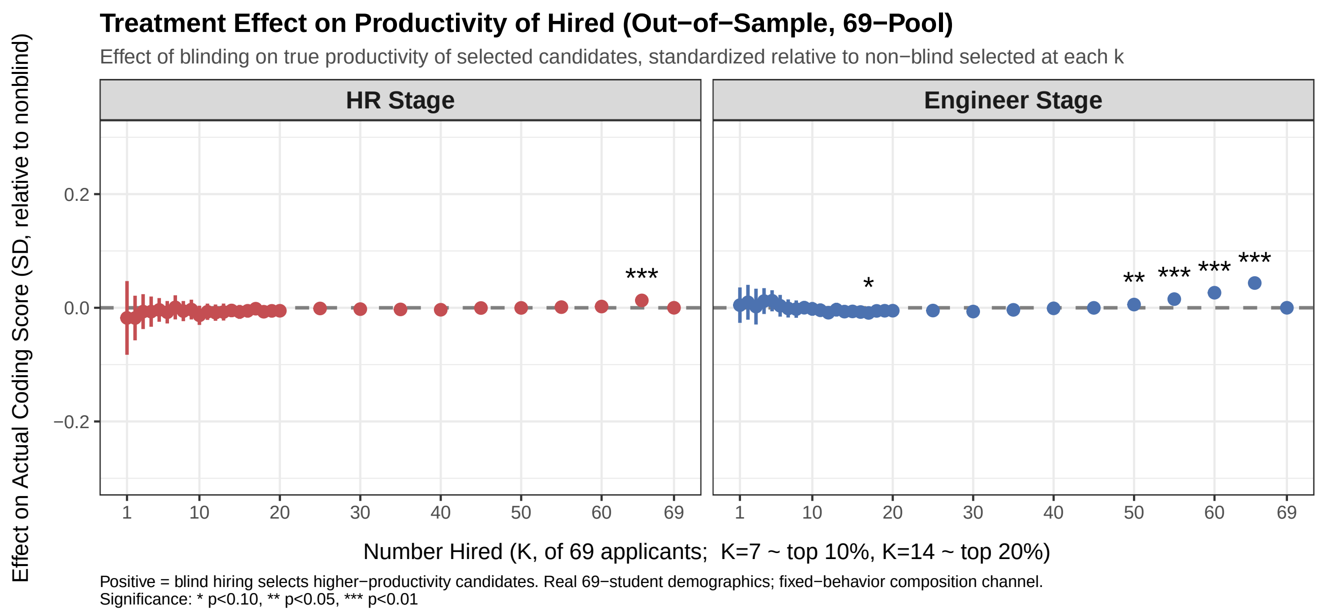
We take the measured non-blind demographic biases and apply them to the true 69-student pool (real demographics, real productivity), simulating blind vs non-blind selection at each hiring cutoff K. Behavior is held fixed (composition channel); this documents how blinding reshapes *who* is hired and whether that moves *productivity*.

1 Compositional changes: who gets hired



Blinding removes the non-blind premia/penalties, so it mechanically reshuffles the demographic composition of hires (e.g., groups that were penalized when identified gain; groups that were given a premium lose). This mirrors the in-sample composition result, now on the real applicant pool.

2 Treatment effect on productivity



Despite the compositional reshuffle, the effect on the true productivity of the hired cohort is null across the entire hiring margin, in both stages. The composition channel moves *who* is hired without moving *how productive* they are — because the biases are modest and demographic identity carries little productivity information beyond the visible signals.

3 Caveats

Calibration, not estimation: behavior held fixed (statistical-discrimination channel deferred); s(X) extrapolated from 18 profiles; bands are simulation-based (evaluator-clustered); the 69 pool is Berkeley-CS (79% Asian, with rare Hispanic-female / White-female cells, so those group panels are noisy).